

Name: _____ Date: _____

Period: _____

Direction of Motion and Slope Check

A particle moves with $f(t) = (2t - t^2, t^2 - 4t + 3)$ for $0 \leq t \leq 4$.

1. At $t = 0.5$, is the particle moving right or left? Up or down? Show your reasoning by comparing values of each component at $t = 0$ and $t = 1$.

$$x(0) = \underline{\hspace{2cm}} \quad x(1) = \underline{\hspace{2cm}} \quad x(t) \text{ is } \underline{\hspace{2cm}} \text{ near } t = 0.5 \Rightarrow \text{moving } \underline{\hspace{2cm}}$$

$$y(0) = \underline{\hspace{2cm}} \quad y(1) = \underline{\hspace{2cm}} \quad y(t) \text{ is } \underline{\hspace{2cm}} \text{ near } t = 0.5 \Rightarrow \text{moving } \underline{\hspace{2cm}}$$

2. Compute the average rate of change of $x(t)$ on $[0, 3]$ and the average rate of change of $y(t)$ on $[0, 3]$. Then find the slope of the curve between the two points.

$$x(0) = \underline{\hspace{2cm}} \quad x(3) = \underline{\hspace{2cm}} \quad \Delta x = \underline{\hspace{2cm}}$$

$$y(0) = \underline{\hspace{2cm}} \quad y(3) = \underline{\hspace{2cm}} \quad \Delta y = \underline{\hspace{2cm}}$$

$$\text{Slope of curve} = \frac{\Delta y}{\Delta x} = \frac{\underline{\hspace{2cm}}}{\underline{\hspace{2cm}}} = \underline{\hspace{2cm}}$$